

Smart Library — 6-Week Development Roadmap

A 6-week plan replacing the previous 2-week sprint, so each pair can develop their area properly instead of constantly switching between unfinished foundational work and advanced features. The core API, database integration, seat map, Flutter functionality, CI/CD, and analyzer cleanup are already substantially completed — this plan builds on that foundation.

Team & Role Assignment

Member	Responsibility
Dhruv	Database + Supabase
Machine Gun Kelly	Backend / API
Yuvraj	Flutter + Web Frontend (Lead)
Subhanshu	Flutter + Web Frontend
Anubhav	Integration, GitHub, PRs, workflow, release
Aman	Minimal documentation
Gaurang	Minimal documentation

Overall Progression

WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5
WEEK 6				
Stabilize	Secure	Improve Seat + Student Exp.	Analytics	Audit + Advanced
Features → Testing + Deployment + Release				

WEEK 1 — Stabilization & Architecture

Machine Gun Kelly — Backend/API

Goal: Stable backend foundation.

- ☐ Review all existing endpoints
- ☐ Standardize API response format
- ☐ Standardize error responses
- ☐ Improve request validation
- ☐ Review /api/scan
- ☐ Review /api/seats
- ☐ Review /api/students-inside
- ☐ Review /api/scan-logs
- ☐ Add centralized error handling
- ☐ Add structured backend logging
- ☐ Document all API dependencies

Dhruv — Database/Supabase

Goal: Reliable database layer. (Hardening, not migrating — already on the team Supabase project with Realtime tables enabled.)

- ☐ Audit all Supabase tables
- ☐ Verify library_config
- ☐ Verify students
- ☐ Verify scan_logs
- ☐ Check relationships and constraints
- ☐ Review indexes
- ☐ Review RLS / security policies
- ☐ Verify timestamp handling
- ☐ Check duplicate / invalid records
- ☐ Prepare a database backup procedure

Yuvraj — Frontend Lead

- ☐ Review Admin Web Panel
- ☐ Review Flutter architecture
- ☐ Standardize navigation
- ☐ Review seat-map integration
- ☐ Fix remaining UI inconsistencies
- ☐ Define common frontend design rules

Subhanshu — Frontend

- ☐ Review Student Portal
- ☐ Review reusable widgets
- ☐ Improve loading states
- ☐ Improve error states
- ☐ Improve empty states
- ☐ Review responsive behavior

Anubhav

- ☐ Establish branch strategy
- ☐ Verify PR process
- ☐ Create integration checklist
- ☐ Ensure nobody directly merges to main
- ☐ Verify development environment for all members

Aman

- ☐ Update root README
- ☐ Update installation instructions

Gaurang

- ☐ Update architecture documentation
 - ☐ Maintain project documentation index
-

WEEK 2 — Authentication, Security & Core Flow

Machine Gun Kelly — Backend

Secure backend flow:

Authentication → Authorization → Validation → Business Logic → Database

- ☐ Improve authentication middleware
- ☐ Add role validation
- ☐ Protect admin routes
- ☐ Add scan rate limiting
- ☐ Improve CORS configuration
- ☐ Handle malformed requests
- ☐ Improve HTTP error codes
- ☐ Add request logging
- ☐ Protect reset endpoint

Dhruv — Database

- ☐ Implement/verify RLS policies
- ☐ Secure student records
- ☐ Secure scan logs
- ☐ Review admin access
- ☐ Optimize scan queries
- ☐ Optimize seat queries
- ☐ Add indexes where required
- ☐ Verify concurrent scan behavior

Yuvraj + Subhanshu — Authentication UI

- ☐ Login
- ☐ Logout
- ☐ Session handling
- ☐ Student/admin routing
- ☐ Unauthorized-state screens
- ☐ Expired-session handling
- ☐ Better login error messages

Anubhav

- ☐ Perform security/integration review
- ☐ Merge verified PRs

Aman + Gaurang

- ☐ Update authentication/API documentation
-

WEEK 3 — Seat Management & Student Experience

Chronological seat allocation and live duration logic are already implemented — this week extends them.

Yuvraj — Admin Web

- ☐ Zone filtering
- ☐ Available/occupied filtering
- ☐ Search student
- ☐ Search seat

- ☐ Seat detail panel
- ☐ Current occupancy
- ☐ Better seat interactions
- ☐ Zone occupancy visualization

Subhanshu — Flutter

- ☐ Current seat
- ☐ Find a seat
- ☐ Seat availability
- ☐ Zone selection
- ☐ Session duration
- ☐ Current library occupancy
- ☐ Library-full warning
- ☐ Better seat details

Machine Gun Kelly — Add/Modify APIs

- ☐ GET /api/seats
- ☐ GET /api/seats/:id
- ☐ GET /api/zones
- ☐ GET /api/occupancy
- ☐ GET /api/students-inside

Dhruv — Optimize Database Queries Supporting

- ☐ Seat allocation
- ☐ Occupancy
- ☐ Zone filtering
- ☐ Student lookup

Anubhav — E2E Validation

Scan → Backend → Database → Seat → Flutter/Web

- ☐ Validate the full scan-to-seat flow end to end

WEEK 4 — Analytics & Admin Management

Machine Gun Kelly — Backend: Analytics APIs

- ☐ GET /api/analytics/daily
- ☐ GET /api/analytics/weekly
- ☐ GET /api/analytics/occupancy
- ☐ GET /api/analytics/zones
- ☐ GET /api/analytics/sessions

Metrics to support: - ☐ Total visits - ☐ Current occupancy - ☐ Peak occupancy - ☐ Average session duration - ☐ Peak hours - ☐ Zone utilization

Dhruv — Database: Analytics Queries/Views

- ☐ Daily visits
- ☐ Weekly visits

- ☐ Monthly visits
- ☐ Peak occupancy
- ☐ Average session duration
- ☐ Zone utilization
- ☐ Ensure raw scan_logs datasets are not loaded directly into the frontend

Yuvraj — Admin Web Analytics

- ☐ Occupancy graph
- ☐ Daily visits
- ☐ Weekly visits
- ☐ Peak hour
- ☐ Zone usage
- ☐ Session duration

Subhanshu — Flutter Analytics

- ☐ Student visit count
- ☐ Personal visit history
- ☐ Average session duration
- ☐ Usage summary

Anubhav — Verify Accuracy Between

Supabase ↔ API Analytics ↔ Web Dashboard ↔ Flutter

- ☐ Confirm analytics numbers match consistently across all four layers

WEEK 5 — Audit System, Notifications & Advanced Features

Machine Gun Kelly — Backend: Admin Audit System

Admin Action → API → Audit Log

Track: - ☐ Student scan - ☐ Force exit - ☐ Reset - ☐ Configuration changes - ☐ Admin actions - ☐ Add notification support where appropriate

Dhruv — Database

- ☐ Create audit_logs table
- ☐ Optimize storage/querying
- ☐ Verify retention strategy
- ☐ Verify indexing
- ☐ Verify data consistency
- ☐ Verify backup/restore process

Yuvraj — Admin Web

- ☐ Audit log
- ☐ Advanced student search
- ☐ Activity filtering
- ☐ Export functionality
- ☐ Improved admin controls

Subhanshu — Flutter

- ☐ Session notifications
 - ☐ Library capacity warnings
 - ☐ Better scan feedback
 - ☐ Current-session alerts
 - ☐ Improved history
-

WEEK 6 — Testing, Deployment & Final Release

Machine Gun Kelly — Backend: Full Testing

Expand the existing backend test suite beyond its current 20/20 validated cases.

- ☐ Invalid requests
- ☐ Duplicate scans
- ☐ Simultaneous scans
- ☐ Full library
- ☐ Student exit
- ☐ Force exit
- ☐ Reset
- ☐ Database failure
- ☐ Authentication failure
- ☐ API failure

Dhruv — Production Database Validation

- ☐ Backup
- ☐ Restore
- ☐ RLS verification
- ☐ Data integrity
- ☐ Migration testing
- ☐ Production configuration
- ☐ Performance checks

Yuvraj + Subhanshu — Complete Frontend QA

Flow to test:

Login → Dashboard → Scan → Seat → Occupancy → History → Profile → Logout

Conditions to test: - ☐ Desktop - ☐ Mobile - ☐ Different browser sizes - ☐ Dark/light mode - ☐ Network failure - ☐ Empty data - ☐ API failure

Note: the Flutter codebase is already at 0 analyzer errors/warnings — this phase should emphasize functional QA and release polish rather than repeating static cleanup.

Anubhav — Release Owner

This becomes his biggest week.

- ☐ Review all open PRs
- ☐ Verify feature branches
- ☐ Run integration checklist
- ☐ Resolve merge conflicts

- ☐ Verify CI/CD
- ☐ Verify Docker
- ☐ Verify environment variables
- ☐ Verify production database
- ☐ Run final E2E test
- ☐ Merge approved PRs
- ☐ Create release tag

Release pipeline (validate, don't rebuild — Docker and GitHub Actions infrastructure already exist):

Feature Branch → Pull Request → Code Review → Tests → Anubhav Verification → Merge → main → v1.0.0

Aman — Documentation Coordinator (Kept Deliberately Small)

No significant coding responsibility — spans all 6 weeks.

- ☐ Update README when features change
- ☐ Update installation/setup instructions
- ☐ Maintain changelog
- ☐ Add screenshots
- ☐ Keep feature list current
- ☐ Maintain release notes

Gaurang — Project Documentation Coordinator (Kept Deliberately Small)

No significant coding responsibility — spans all 6 weeks.

- ☐ Update architecture diagrams
- ☐ Maintain API documentation
- ☐ Maintain database schema documentation
- ☐ Update UML diagrams
- ☐ Maintain project report material
- ☐ Maintain meeting/progress documentation

Final Responsibility Matrix

Area	Primary	Secondary
Backend/API	Machine Gun Kelly	Anubhav
Database/Supabase	Dhruv	Machine Gun Kelly
Flutter	Yuvraj	Subhanshu
Web Frontend	Yuvraj	Subhanshu
Integration	Anubhav	Everyone
GitHub/PR/Merging	Anubhav	—
Documentation	Aman	Gaurang
API Docs	Aman	Gaurang
DB Docs	Gaurang	Dhruv

Testing	Machine Gun Kelly	Anubhav
Release	Anubhav	Machine Gun Kelly
UI/UX	Yuvraj	Subhanshu